

Dmytro Humeniuk

AI Software Developer | Robotics, Computer Vision & MLOPs

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Summary

AI Software Developer with a PhD in Computer Engineering and more than five years of combined industry and research experience developing production-oriented AI and computer vision solutions for autonomous robotic systems. Strong hands-on experience building end-to-end perception pipelines, including data generation and processing, model training and evaluation, robotic integration, deployment, and automated testing. Experienced in deep-learning-based object detection, perception-guided robotic manipulation, synthetic data generation, sim-to-real transfer, and deployment of AI components to edge computing platforms. Strong Python and C++ programming skills with experience designing modular software architectures, APIs, CI/CD workflows, and reproducible ML pipelines. Proven ability to integrate perception models with real-world robotic systems and collaborate across AI, software and hardware teams.

Skills

Programming: Python, C/C++, SQL, Shell

Machine Learning & AI: PyTorch, TensorFlow, Scikit-Learn, Gymnasium, OpenCV, TensorRT, Hailo

Robotics & Simulation: ROS 2, NVIDIA Isaac Sim, Gazebo, Ardupilot

Cloud & DevOps: Docker, Kubernetes, AWS, Azure, Google Cloud, CI/CD, Weights&Biases

Development Tools: Git, Flask, Linux, gRPC, REST APIs

Languages: English (fluent), French (fluent), Ukrainian (native), Russian (native)

Experience

Autonomous Systems Developer, [SkyReach Aerospace](#), Montréal

July 2026 – present

- Developed image-based visual servoing control for unmanned aerial vehicles.
- Implemented real-time object detection and optical flow based tracking algorithms.
- Designed software and hardware architecture for the autonomy stack.

Artificial Intelligence Developer, Thales Digital Solutions, Montréal

Sept 2025 – May 2026

- Developed production-oriented Python software and AI components for collaborative autonomy and robotic-system simulation.
- Implemented perception-based obstacle-avoidance algorithms for autonomous robotic systems.
- Developed gRPC-based APIs and deployed AI software to edge computing platforms.

Research Intern, Sycodal, Montréal

May 2023 – Sept 2025

- Developed an end-to-end Python framework for training, testing, and validating computer vision and robotic-perception models in NVIDIA Isaac Sim under varying environmental and operating conditions.
- Integrated a deep-learning YOLO-based object detector with robot motion-planning and control components to support automated 6D pose estimation and pick-and-place execution with Kinova Link 6 and Flexiv Rizon 4s robotic manipulators.
- Trained reinforcement learning-based policies for robotic manipulator control in Isaac Lab.
- Domain adaptation of generated synthetic image data in Isaac Sim using generative style transfer models (CycleGANs, CUT).
- Authored a paper published at the [IEEE/ACM International Conference on Automated Software Engineering](#).

Teaching Assistant – Advanced Software Testing, Polytechnique Montréal

Jan 2024 – May 2024

- Designed and implemented five assignments covering fuzzing, search-based software testing, and large language model-based test generation. Presented lab sessions and graded assignments for a group of 12 students.

Machine Learning Software Developer Intern, Ericsson, Montréal

Jan 2023 – Jun 2023

- Implemented decentralized and distributed gradient-descent algorithms in Python using PyTorch and BlueFog.
- Designed controlled experiments to measure communication-computation trade-offs across different network topologies in a real-world 10-node computing environment.

Research Intern, COGECO, Montréal

Feb 2021 – Jun 2021

- Collected, cleaned, analyzed, and visualized large-scale preventive-maintenance data from cable modem networks using Python. Performed feature extraction, feature engineering, statistical analysis, and dimensionality assessment to identify indicators of equipment degradation.
- Developed and evaluated a predictive-maintenance model for forecasting equipment failures up to seven days in advance.

Selected Projects

- **Dynasto** – Developed an open-source tool using reinforcement learning and search-based optimization to generate adversarial driving behaviours, increasing performance over baselines by 20%. Obtained distinguished paper award at ICST 2026 conference. Code: [Dynasto](#). Paper: [Dynasto](#).
- **VehicleDetector** – Developed an open-source computer vision pipeline covering data annotation, model training, validation, and performance evaluation for vehicle detection. Code: [VehicleDetector](#).
- **RILaST** – Developed a variational autoencoder-based Python framework using latent-space optimization to improve failure discovery in autonomous vehicles and drones, 3 to 4.6 times more effective than baseline approaches. Code: [RILaST](#). Paper: [RILaST](#).
- **RIGAA** – Combined reinforcement learning and evolutionary optimization for automated generation of safety-critical test scenarios outperforming the baselines by 20 to 37 %. Code: [RIGAA](#). Paper: [RIGAA](#).
- **Kinova Gen3 digital twin** – Developed ROS2 based digital twin of a real-world Kinova Gen3 robotic manipulator. Demo: [KinovaDigitalTwin](#).

Education

PhD in Computer Engineering, with distinction, Polytechnique Montréal

Sep 2021 – Dec 2025

Thesis: [Operational Knowledge Distillation for Efficient Testing and Enhancement of Autonomous Robotic Systems](#), Supervisor: [Prof. Foutse Khomh](#)

Master of Applied Science in Computer Engineering, Polytechnique Montréal

Sep 2019 – Aug 2021

Thesis: [A Search-Based Framework for Automatic Generation of Testing Environments for Cyber-Physical Systems](#), Supervisor: [Prof. Foutse Khomh](#)

Bachelor Degree in Computer Engineering, Kyiv Polytechnic Institute

Sep 2015 – Jun 2019

Final Project: [Automated System for Determining Solar Cell Model Parameters](#)

Scholarships and Awards

- Merit Scholarship for Foreign Students, PBEEE, 50,000 CAD May 2024
- FRQNT Doctoral Training Scholarship May 2024
- CodeML Hackathon – 2nd place, CNN-based model for plant disease prediction Oct 2023
- CodeML Hackathon – 1st place, NLP model for language classification Oct 2022
- Finalist, [Human Competitive Awards](#), GECCO Conference July 2022
- Winner, [SBST CPS Testing Competition](#), AmbieGen tool June 2022

Other Activities

- Member of the [organizing committee](#) of the international competition on autonomous drone testing 2024 – 2026
- [Participant](#), Montreal Summer School in Robotics at Mila Aug 2022
- [Participant](#), IVADO/Mila Deep Learning School Mar 2021

Volunteer Activities

- Volunteer for an automated de-mining system project. Development of computer vision models for land mine detection, dataset collection and processing. Feb – Sept 2025
- Member of the organizing committee for the [AI Helps Ukraine conference](#). June – Dec 2022
- Volunteer for the Ukrainian community; organized [fundraising](#) for a VAC device for a hospital in Kyiv, raising 3,000 CAD; Feb – Dec 2022